

PBMC3k Single-Cell RNA-seq Downstream Analysis Report

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Report Title

PBMC3k Single-Cell RNA-seq Downstream Analysis Report

Conversation Name

Analysis Objective: Perform standard downstream analysis on 10x Genomics PBMC3k...

Report Time

2026-09-04 18:26

1. Results

1.1 Overview

This report summarizes the downstream analysis of the 10x Genomics PBMC3k single-cell RNA-seq dataset. The analysis pipeline comprised quality control (QC), normalization, dimensionality reduction (PCA and UMAP), graph-based clustering, cluster marker identification, and cell-type annotation using classical PBMC marker genes. All analyses were performed with Seurat 5.5.1 in R 4.5.2.

1.2 Quality Control

The 10x filtered gene-barcode matrix was loaded with Seurat Read10X, yielding 32,738 genes across 2,700 cells with 2,286,884 non-zero entries. Thirteen mitochondrial genes (^MT-) were detected. Per-cell QC metrics (nFeature_RNA, nCount_RNA, percent.mt) were computed. Cells with fewer than 200 detected genes (too few genes) or with mitochondrial proportion $\geq 5\%$ (excessively high) were removed as low quality.

Metric	Value	Notes
input_cells	2700	
input_genes	32738	
nonzero_entries	2286884	
mitochondrial_genes_detected	13	
cells_kept_after_qc	2643	
cells_removed_by_qc	57	

QC violin plots (before and after filtering) are shown in Figure 1. Per-cell QC metrics including filter status were written to outputs/files/qc_metrics.csv.

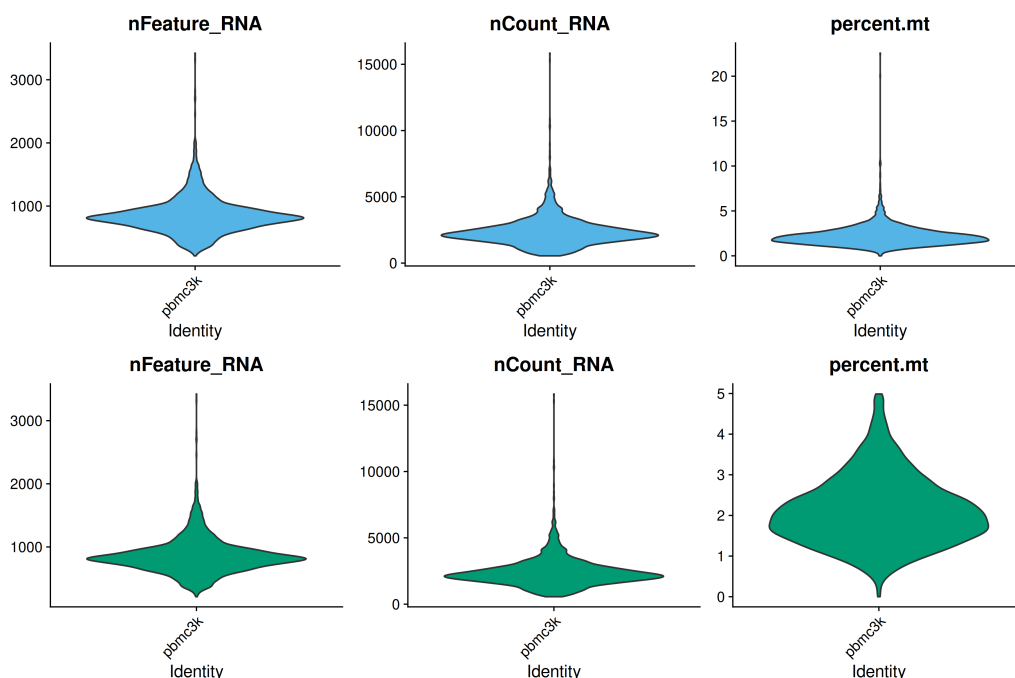


Figure 1 QC violin plots before and after filtering (nFeature_RNA, nCount_RNA, percent.mt).

The violin plots show the distribution of QC metrics before and after filtering. After filtering, cells with very low gene counts and high mitochondrial proportions are removed, resulting in a cleaner distribution.

1.3 Normalization, Dimensionality Reduction, and Clustering

The filtered Seurat object (2,643 cells x 32,738 genes) was normalized with LogNormalize (scale.factor=10,000). The top 2,000 highly variable genes (HVGs) were selected using the VST method and scaled. PCA was run over 50 PCs; an elbow heuristic selected 9 significant PCs (51.73% cumulative variance). Graph-based Louvain clustering (algorithm=1, resolution=0.5, random.seed=42) on the 9-PC neighbor graph produced 10 clusters (0-9). UMAP was computed with the uwot implementation (cosine metric, dims=1:9).

Cluster	Cells
0	548
1	486
2	477
3	344
4	286
5	164
6	148
7	143
8	33
9	14

The PCA plot colored by cluster is shown in Figure 2, and the UMAP plot colored by cluster with labels is shown in Figure 3.

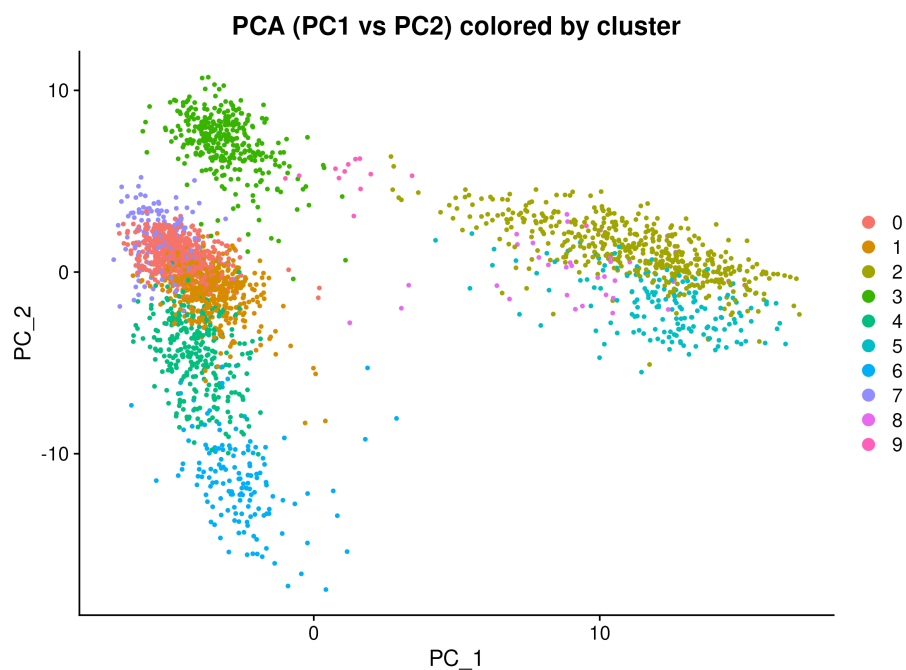


Figure 2 PCA plot colored by cluster.

The PCA plot shows the first two principal components colored by Louvain cluster. The clusters are partially separated in PCA space, reflecting the major axes of variation in the data.

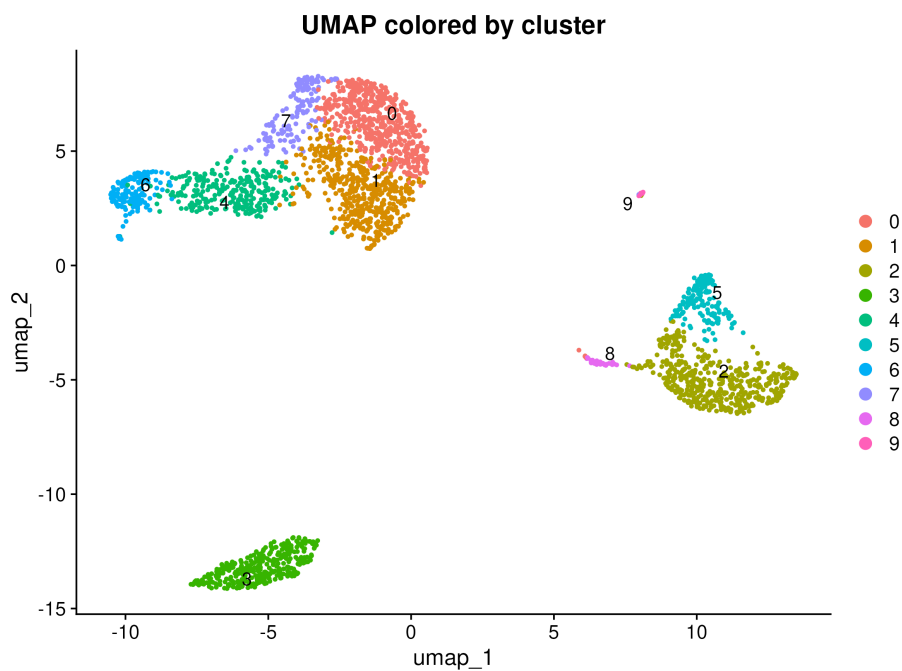


Figure 3 UMAP plot colored by cluster with labels.

The UMAP plot shows 10 distinct clusters (0-9) with clear separation between most clusters. Cluster sizes range from 14 to 548 cells.

1.4 Cluster Marker Genes

FindAllMarkers (Wilcoxon rank-sum test, only.pos=TRUE, min.pct=0.25, logfc.threshold=0.25) identified 3,527 positive marker rows across all 10 clusters, corresponding to 2,026 unique genes. The full marker table was saved to outputs/files/cluster_markers.csv.

Metric	Value	Notes
marker_rows_found	3527	
unique_marker_genes	2026	

1.5 Cell Type Annotation

Clusters were annotated to 9 major PBMC cell types using classical marker-gene scoring (cluster-level mean z-score of marker panels) cross-checked against top differential markers. Cluster 7 was refined to CD8 T based on a cytotoxic signature (GZMK/CCL5/NKG7) with CD8A>CD4 expression.

Cluster	Cell Type	Key Markers
0	Naive CD4 T	IL7R, CCR7, LEF1, TCF7
1	Memory CD4 T	IL7R, S100A4, CD40LG
2	CD14 Mono	CD14, LYZ, S100A9
3	B cells	MS4A1, CD79A
4	CD8 T	CD8A, GZMK, CCL5
5	FCGR3A Mono	FCGR3A, MS4A7
6	NK	GNLY, NKG7, GZMB

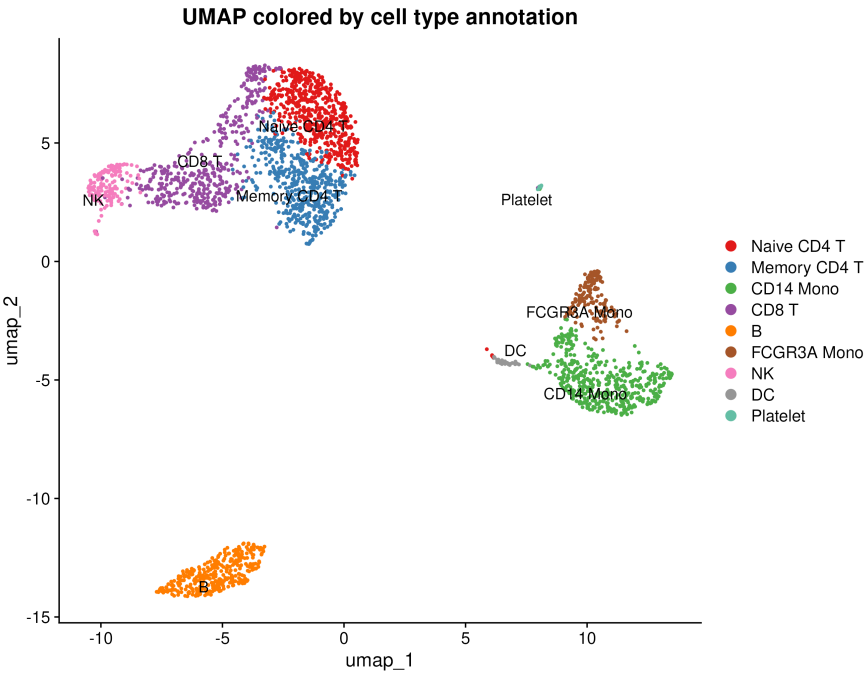
Cluster	Cell Type	Key Markers
7	CD8 T	CD8A, GZMK, CCL5, NKG7
8	DC	FCER1A, CD1C, CST3
9	Platelet	PPBP, PF4, ITGA2B

Metric	Value	Notes
cells_annotated	2643	
num_clusters_annotated	10	
num_cell_types	9	

Cell type counts and proportions are summarized below and saved to outputs/files/celltype_counts.csv.

Cell Type	Count	Proportion
Naive CD4 T	548	20.7%
Memory CD4 T	486	18.4%
CD14 Mono	477	18.0%
CD8 T	429	16.2%
B	344	13.0%
FCGR3A Mono	164	6.2%
NK	148	5.6%
DC	33	1.3%
Platelet	14	0.5%

The UMAP plot colored by cell type annotation is shown in Figure 4, and the classical marker gene expression heatmap is shown in Figure 5.



The UMAP plot colored by cell type shows 9 major PBMC cell types. T cell subsets (Naive CD4, Memory CD4, CD8) and monocyte subsets (CD14, FCGR3A) are separated, along with B cells, NK cells, DCs, and platelets.

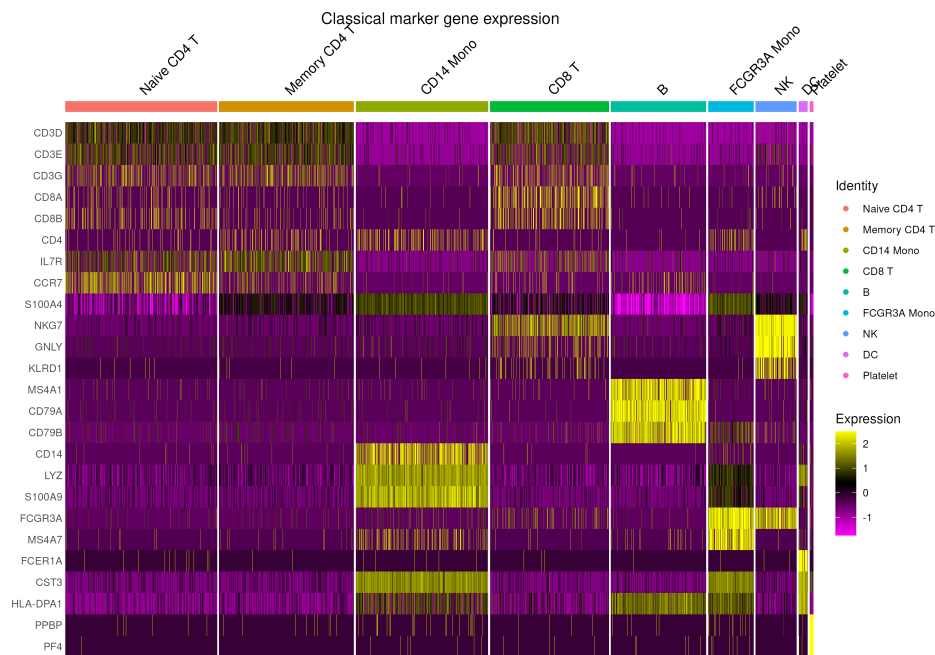


Figure 5 Heatmap of classical marker gene expression across cell types.

The heatmap shows expression of 25 classical marker genes across the 9 annotated cell types. Each cell type shows a distinct marker expression profile, confirming the annotation.

1.6 Methods and Parameters

- Software: Seurat 5.5.1, R 4.5.2
- Input: 10x filtered gene-barcode matrix (barcodes.tsv, genes.tsv, matrix.mtx)
- QC metrics: nFeature_RNA, nCount_RNA, percent.mt (^MT- pattern)
- Filtering: nFeature_RNA >= 200 and percent.mt < 5
- Normalization: LogNormalize, scale.factor=10000
- HVG selection: VST, nfeatures=2000
- PCA: 50 PCs, seed 42; significant PCs via elbow heuristic (max chord distance) -> 9 PCs
- Clustering: Louvain (algorithm=1), resolution=0.5, random.seed=42
- UMAP: uwot, cosine metric, dims=1:9, seed 42
- Marker identification: Wilcoxon rank-sum, only.pos=TRUE, min.pct=0.25, logfc.threshold=0.25
- Annotation: classical PBMC marker panel, cluster-level z-score scoring (argmax per cluster), with cytotoxic CD8 T refinement rule (GZMK>1.5 and CCL5>8 avg expr)

Final Answer: The PBMC3k single-cell RNA-seq downstream analysis was completed successfully. After QC filtering (nFeature_RNA >= 200, percent.mt < 5), 2,643 of 2,700 cells were retained. Normalization (LogNormalize), HVG selection (2,000 genes), PCA (9 significant PCs), Louvain clustering (resolution 0.5) yielded 10 clusters, and UMAP was computed. FindAllMarkers identified 3,527 marker rows (2,026 unique genes). Clusters were annotated to 9 major PBMC cell types using classical marker genes: Naive CD4 T (548), Memory CD4 T (486), CD14 Mono (477), CD8 T (429), B (344), FCGR3A Mono (164), NK (148), DC (33), and Platelet (14). All required output files were generated, including QC metrics, violin plots, PCA plot, UMAP cluster plot, cluster markers, UMAP cell-type plot, cell type counts, and marker gene heatmap.